

Charlene Khun

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PROJECTS

Automobile Efficiency Narrative Visualization | Stack: D3.js, JavaScript, HTML, CSS Aug 2026

- Developed an interactive D3.js visualization to analyze relationships among engine cylinders, city/highway fuel efficiency, and fuel type using a 2017 automobile dataset.
- Performed data parsing, validation, aggregation, and linear regression analysis, presenting findings through scatter plots, grouped bar charts, annotations, and interactive tooltips.
- Built dynamic filtering and state-driven navigation for gasoline, diesel, and electric vehicle comparisons, transforming raw automobile data into an interactive three-scene analytical story.

COVID-19 Trends & Geographic Analysis Dashboard | Stack: Tableau June 2026

- Developed an interactive Tableau dashboard analyzing state-level U.S. COVID-19 case and death trends using geographic and time-series visualizations.
- Built choropleth and dual-axis line charts with interactive tooltips and bi-directional cross-filtering to support state- and date-level analysis.
- Transformed raw public health data into an intuitive analytical dashboard that enables users to identify geographic patterns, compare temporal trends, and investigate changes in cases and deaths.

SF Rent Analysis & Prediction | Stack: Python, Pandas, Seaborn, scikit-learn April. 2025 – May 2025

- Cleaned and transformed San Francisco rental housing data by handling missing values, standardizing data types, removing irrelevant fields, and engineering new features for analysis.
- Conducted exploratory analysis using correlation matrices, histograms, pair plots, and choropleth maps to identify neighborhood trends and key factors associated with monthly rent.
- Evaluated regression models using RMSE, MAE, and R^2 , identifying Random Forest as the best-performing model and using feature importance analysis to interpret major drivers of rental prices.

NYPL Data Cleaning & Analysis Project | Stack: OpenRefine, SQL, Python/Pandas June 2026 - Aug 2026

- Profiled and cleaned historical restaurant-menu data using OpenRefine, identifying missing values, invalid dates, inconsistent currencies, pricing anomalies, and unnecessary attributes across four relational datasets.
- Prepared 1.3M+ menu-item records for analysis by standardizing data types, filtering records to comparable historical periods and currencies, and preserving valid relational keys for downstream analysis.
- Collaborated on SQL-based data validation and integrity constraints and documented transformation histories to create a reproducible pipeline for reliable historical menu-price analysis.

EDUCATION

University of Illinois Urbana - Champaign

Master of Computer Science (MCS)

- GPA: 3.89/4.0

San Jose State University

Bachelor of Science in Computer Science (BSCS)

- GPA: 3.783/4.0, Cum Laude

Dec 2026

Remote

May 2025

San Jose, CA

CERTIFICATIONS, SKILLS & INTERESTS

- **Certifications:** UMich PY4E Specialization, Flask, HTML/CSS, Tailwind, Bootstrap, Google Data Analytics (In Progress), Google Advanced Data Analytics (In Progress)
- **Technologies:** Python, SQL, Tableau, D3.js, JavaScript, HTML/CSS, OpenRefine, Github, AWS
- **Skills:** Data Cleaning, EDA, Data Visualization, Data Storytelling, Statistical Analysis
- **Interests:** Foodie, Travelling, Music, Piano